Logistic Regression Modeling

Predicting Hospital Readmission from Patient Variables

Research Question

Which variables have the most significant effect on whether or not a patient will be re-admitted after initial treatment?

Analysis Goals

This data analysis aims to build a refined logistic model able to determine whether a patient will need to be readmitted by selecting the most significant independent variables from the data set from an initial logistic model.

Logistic Regression Assumptions

- Logistic regression assumes that the model's dependent variable is binary.
- It also assumes that there is a linear relationship between the explanatory variables and the logit function.
- Logistic regression also assumes no multicollinearity between the independent variables.
- Finally, it is assumed that all observations from our medical dataset are independent of each other.

Tool & Technique Justification

Python will be used for this data analysis because of the packages available, such as seaborn and matplotlib for visualization. Additionally, the ease of panda's capability to transform the data file into a dataframe and work with the data is extremely helpful, particularly for building and operating within dummy dataframes.

Logistic regression will be the appropriate technique because the dependent variable to be tested is a binary categorical variable and the goal of the data analysis is a binary yes/no recommendation.

Data Cleaning

First, the data has many yes/no columns. Data analysis will be much easier if these are converted to 1/0. This is achieved by running a .replace function over the entire dataframe converting yes/no to 1/0. This is done after creating a dummies dataframe to ensure to the dummy values are also one hot encoded.

Duplicates will be evaluated using .duplicate function, but no fields populate that are necessary to change.

Outlier analysis is also performed on all numeric columns to be used in the initial logistic model, and outliers are replaced with the column mean. There are no columns with outliers however, so no changes are made beyond the fill NA function added as a just in case.

All categorical columns are evaluated using .unique to ensure there are no duplicates, nulls, or values in need of conversion, and these all look good to proceed.

Summary Statistics

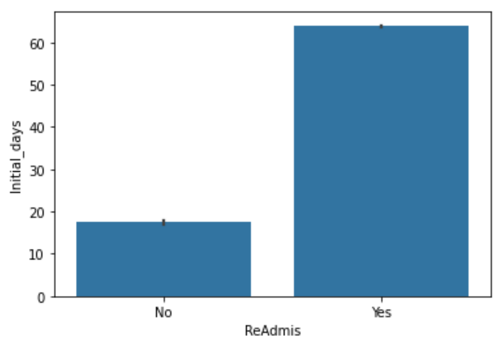
The summary statistics show that readmission is most often not necessary, with 6331 non re-admitted patient interactions out of the 10,000 total interactions. The average patient is 53.5 years old with a max of 89 years. There are 5018 females in the dataset. Emergency is the most common admission reason with 5060 such occurrences. Most patients marked that they do not experience anxiety, at 6785 such responses. The average patient stays 34.4 days in the hospital, and the longest stay is just under 72 full days. We are able to see that 58.65% of patients do not have reflux esophagitis. The most commonly used service is bloodwork, with 5265 patients needing it. 66.28% of patients do not have hyperlipidemia. 72.62% of patients are free of diabetes. 70.94% of patients do not classify as overweight. 80.07% of patients have not experienced a stroke prior to admission. 59.10% of patients do not have high blood pressure.

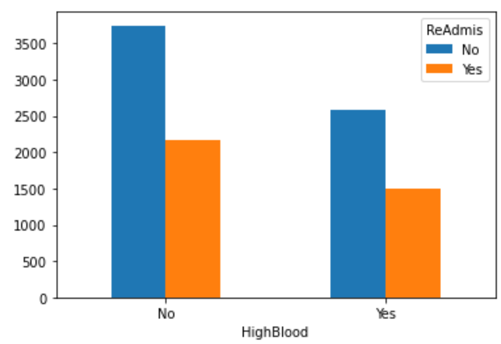
```
Name: count, dtype: int64
count     10000
unique        2
top          No
freq       6331
Name: ReAdmis, dtype: object
count    10000.000000
mean        53.511700
std         20.638538
min         18.000000
25%         36.000000
50%         53.000000
75%         71.000000
max         89.000000
Name: Age, dtype: float64
count      10000
unique         3
top       Female
freq        5018
Name: Gender, dtype: object
count                     10000
unique                        3
top       Emergency Admission
freq                       5060
Name: Initial_admin, dtype: object
count     10000
unique        2
top          No
freq       6785
Name: Anxiety, dtype: object
```

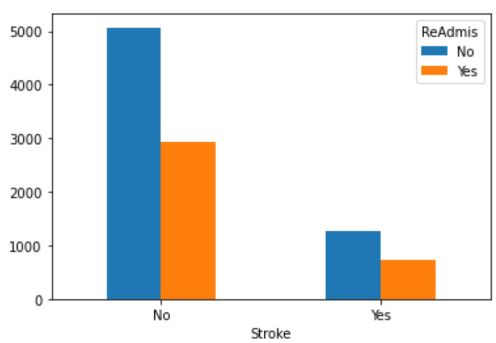
```
count    10000.000000
mean        34.455299
std         26.309341
min          1.001981
25%          7.896215
50%         35.836244
75%         61.161020
max         71.981490
Name: Initial_days, dtype: float64
count     10000
unique        2
top          No
freq       5865
Name: Reflux_esophagitis, dtype: object
count          10000
unique             4
top       Blood Work
freq            5265
Name: Services, dtype: object
count     10000
unique        2
top          No
freq       6628
Name: Hyperlipidemia, dtype: object
count     10000
unique        2
top          No
freq       7262
Name: Diabetes, dtype: object
count     10000
unique        2
top         Yes
freq       7094
Name: Overweight, dtype: object
```

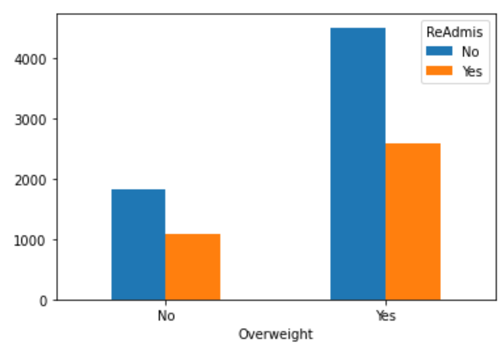
```
Name: Hyperlipidemia, dtype: object
count      10000
unique         2
top           No
freq        7262
Name: Diabetes, dtype: object
count      10000
unique         2
top          Yes
freq        7094
Name: Overweight, dtype: object
count      10000
unique         2
top           No
freq        8007
Name: Stroke, dtype: object
count      10000
unique         2
top           No
freq        5910
Name: HighBlood, dtype: object
C:\Users\welez\AppData\Local\Temp\ipykernel_1
```

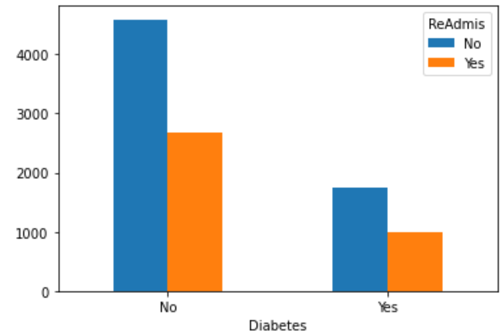
Univariate & Bivariate Visualizations

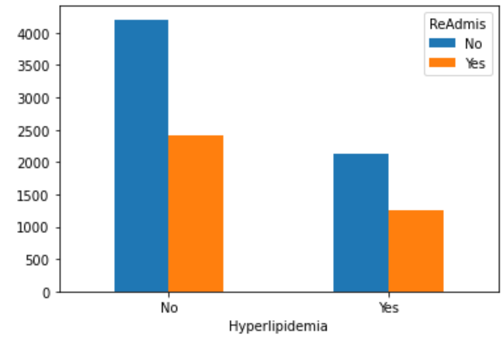


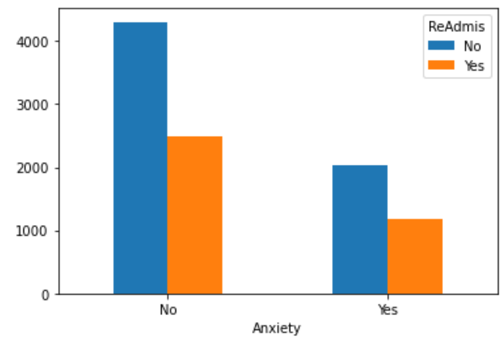


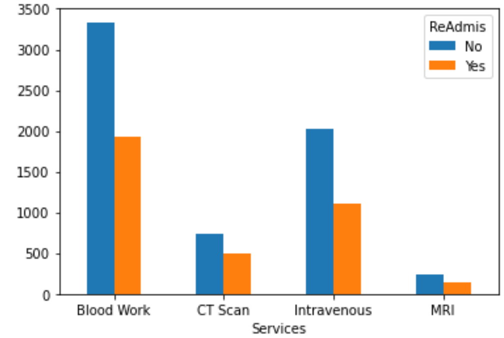


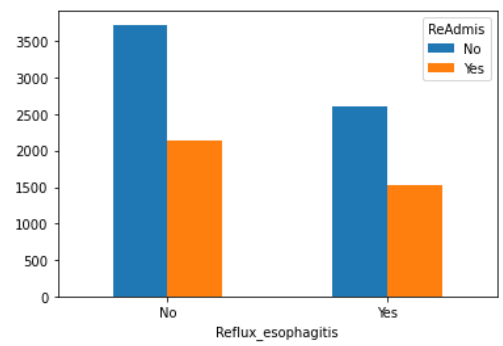


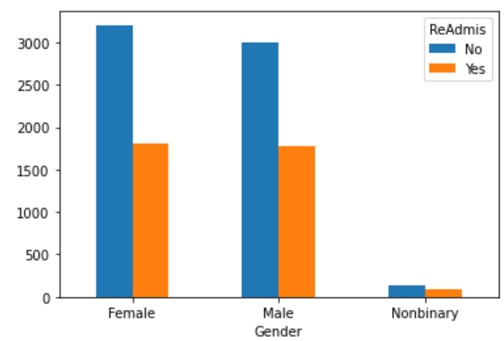

ReAdmis
No
Yes
0
1000
2000
3000
4000
No
Yes
Diabetes

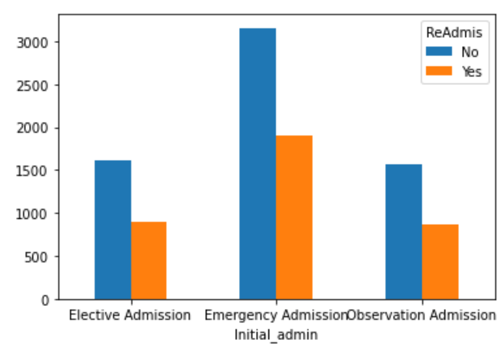

ReAdmis
No
Yes
0
500
1000
1500
2000
2500
3000
3500
4000
No
Yes
Hyperlipidemia

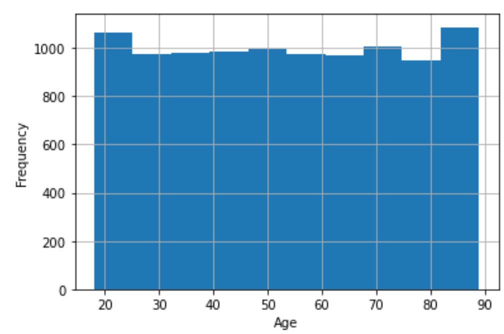

ReAdmis
No
Yes
0
1000
2000
3000
4000
No
Yes
Anxiety


ReAdmis
No
Yes
0
500
1000
1500
2000
2500
3000
3500
Blood Work
CT Scan
Intravenous
MRI
Services


ReAdmis
No
Yes
0
500
1000
1500
2000
2500
3000
3500
No
Yes
Reflux_esophagitis


ReAdmis
No
Yes
0
500
1000
1500
2000
2500
3000
Female
Male
Nonbinary
Gender


ReAdmis
No
Yes
0
500
1000
1500
2000
2500
3000
Elective Admission
Emergency Admission
Observation Admission
Initial_admin


Frequency
0
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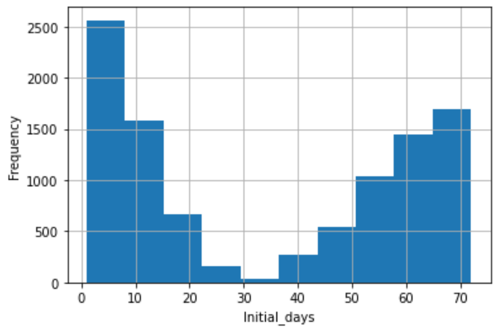
400
600
800
1000
20
30
40
50
60
70
80
90
Age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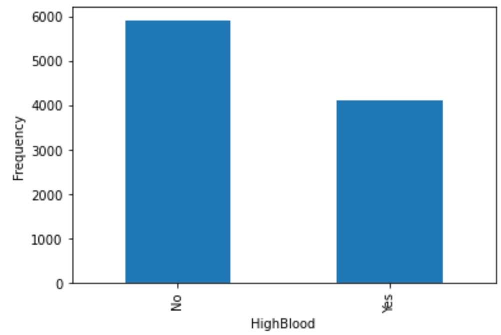
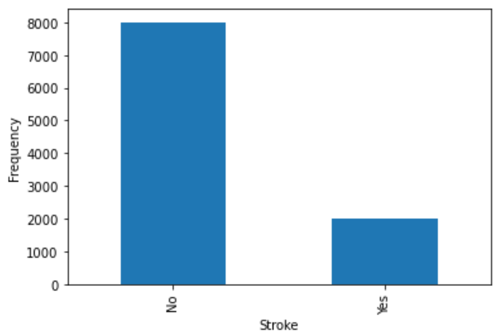
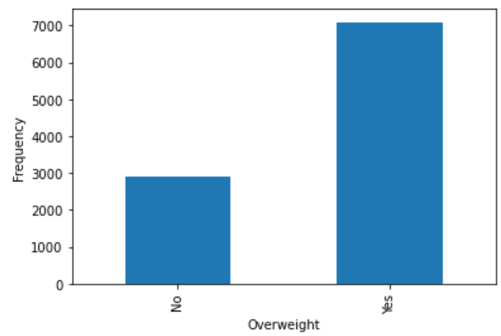
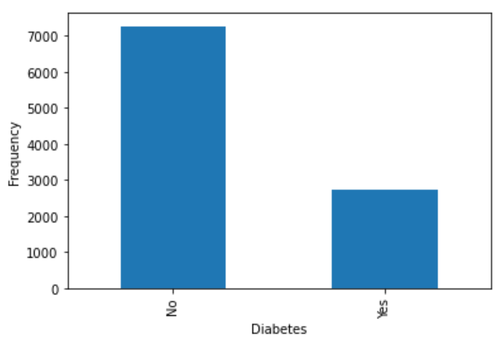
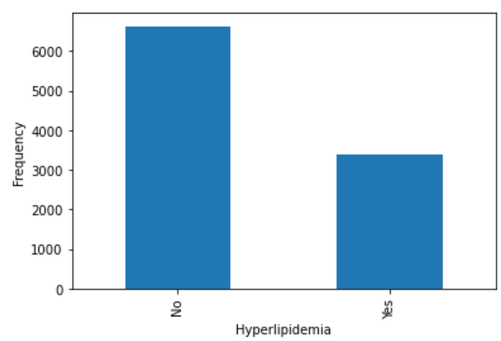
2000
1500
1000
500
0
0 10 20 30 40 50 60 70
Initial_days


Frequency
6000
5000
4000
3000
2000
1000
0
No
Yes
HighBlood


Frequency
8000
7000
6000
5000
4000
3000
2000
1000
0
No
Yes
Stroke


Frequency
7000
6000
5000
4000
3000
2000
1000
0
No
Yes
Overweight


Frequency
7000
6000
5000
4000
3000
2000
1000
0
No
Yes
Diabetes


Frequency
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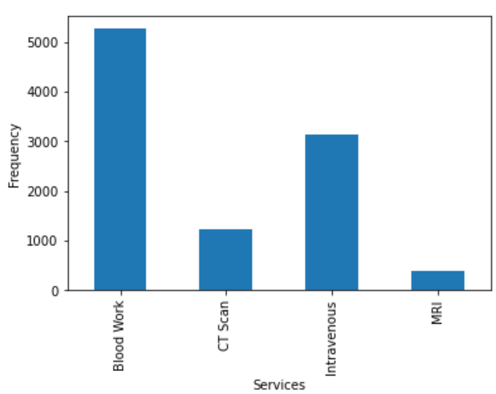
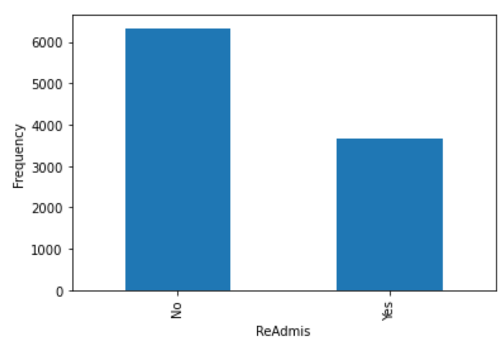
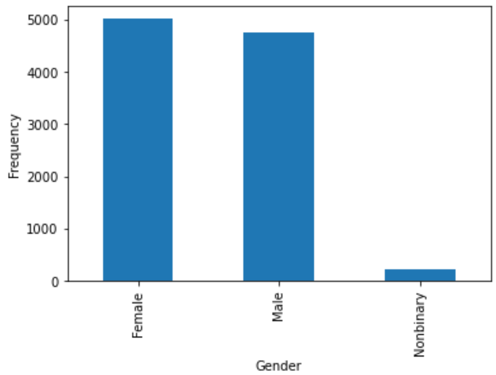
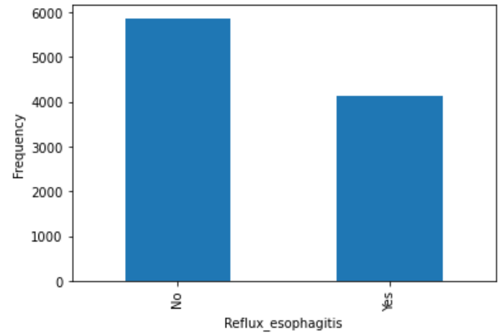
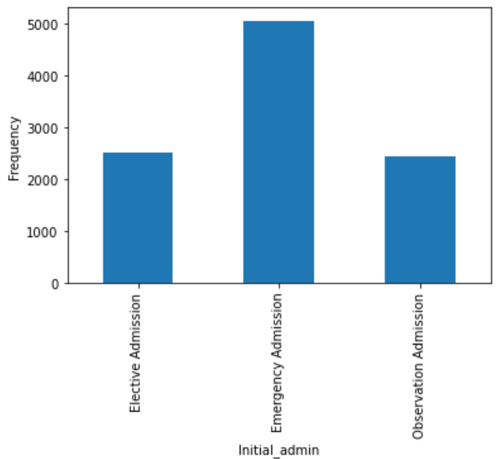
5000
4000
3000
2000
1000
0
No
Yes
Hyperlipidemia

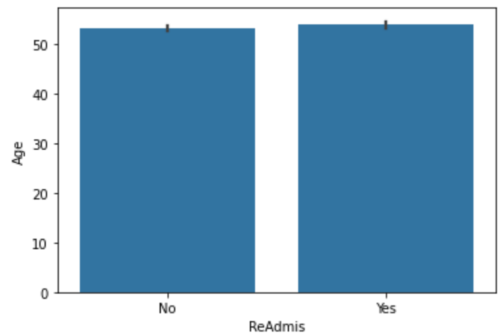

Frequency
5000
4000
3000
2000
1000
0
Blood Work
CT Scan
Intravenous
MRI
Services

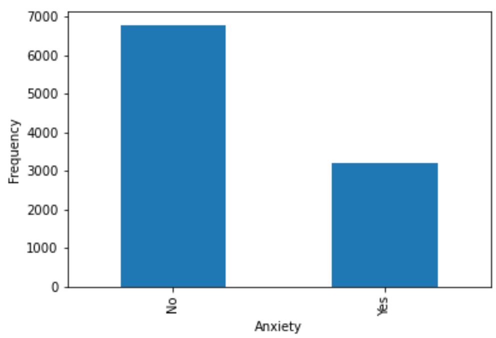

Frequency
6000
5000
4000
3000
2000
1000
0
No
Yes
ReAdmis











Data Transformation

The categorical columns will need to be transformed into dummy columns and built into a new dataframe. This will be done using pandas' get dummies function and assigning these values to a dataframe that will be evaluated in feature selection.

The dummy variables will need to be one hot encoded, converting any True/False and Yes/No columns into 1/0s so we can run calculations.

The data will then be split into a training and test group, using an 80/20 split.

Then, the data will be normalized using the minmaxscaler.

Finally, backwards RFE will be implemented on the logistic model variables until ten features remain for the reduced model.

Initial Model

```
<class 'statsmodels.iolib.summary.Summary'>
"""
                           Logit Regression Results
==============================================================================
Dep. Variable:                      y   No. Observations:                 8000
Model:                          Logit   Df Residuals:                     7984
Method:                           MLE   Df Model:                           15
Date:                Tue, 15 Apr 2025   Pseudo R-squ.:                  0.6019
Time:                        14:33:15   Log-Likelihood:                -2098.5
converged:                       True   LL-Null:                       -5271.6
Covariance Type:            nonrobust   LLR p-value:                     0.000
=========================================================================================================
                                          coef    std err          z      P>|z|      [0.025      0.975]
---------------------------------------------------------------------------------------------------------
Age                                    -2.4447      0.134    -18.267      0.000      -2.707      -2.182
Initial_days                            8.3249      0.184     45.225      0.000       7.964       8.686
 Reflux_esophagitis_Yes                -0.8004      0.080    -10.014      0.000      -0.957      -0.644
Gender_Male                            -0.8199      0.080    -10.307      0.000      -0.976      -0.664
Gender_Nonbinary                       -0.4506      0.267     -1.686      0.092      -0.974       0.073
Anxiety_Yes                            -0.6407      0.084     -7.621      0.000      -0.805      -0.476
Initial_admin_Emergency:Admission      -1.3092      0.096    -13.638      0.000      -1.497      -1.121
Initial_admin_Observation:Admission    -1.5455      0.112    -13.828      0.000      -1.765      -1.326
Services_CT_Scan                       -0.3398      0.128     -2.665      0.008      -0.590      -0.090
Services_Intravenous                   -0.7511      0.088     -8.566      0.000      -0.923      -0.579
Services_MRI                           -0.3161      0.204     -1.550      0.121      -0.716       0.084
Stroke_Yes                             -0.4099      0.097     -4.232      0.000      -0.600      -0.220
Hyperlipidemia_Yes                     -0.5559      0.083     -6.731      0.000      -0.718      -0.394
Diabetes_Yes                           -0.6421      0.088     -7.291      0.000      -0.815      -0.469
Overweight_Yes                         -1.3978      0.089    -15.786      0.000      -1.571      -1.224
HighBlood_Yes                          -0.6005      0.080     -7.512      0.000      -0.757      -0.444
=========================================================================================================
"""
```

Feature Selection

Recursive backward feature elimination will be implemented in order to continually analyze the least important feature from the initial model and remove it, repeating continuously until only ten features remain. This will ensure that any changes to the remaining variables are accounted for upon each step of removal, and that will occur as many features are directly related.

```
    ...: print(cm)
Initial Y Intercept           : [-17.01035867]
[[3.60713838e-02 9.63928616e-01]
 [9.99990535e-01 9.46454581e-06]
 [9.99999903e-01 9.73228730e-08]
 ...
 [4.16635088e-01 5.83364912e-01]
 [9.99981715e-01 1.82847730e-05]
 [6.00332271e-01 3.99667729e-01]]
              precision    recall  f1-score   support

           0       0.99      0.97      0.98      5040
           1       0.95      0.99      0.97      2960

    accuracy                           0.98      8000
   macro avg       0.97      0.98      0.98      8000
weighted avg       0.98      0.98      0.98      8000

[[4890  150]
 [  25 2935]]
```

Reduced Model

```
<class 'statsmodels.iolib.summary.Summary'>
"""
                           Logit Regression Results
==============================================================================
Dep. Variable:                      y   No. Observations:                 8000
Model:                          Logit   Df Residuals:                     7990
Method:                           MLE   Df Model:                            9
Date:                Tue, 15 Apr 2025   Pseudo R-squ.:                  0.3721
Time:                        13:14:10   Log-Likelihood:                -3309.9
converged:                       True   LL-Null:                       -5271.6
Covariance Type:            nonrobust   LLR p-value:                     0.000
==============================================================================================
                                 coef    std err          z      P>|z|      [0.025      0.975]
----------------------------------------------------------------------------------------------
Initial_days                   4.7413      0.102     46.526      0.000       4.542       4.941
Initial_admin_Emergency Admission  -1.2817      0.060    -21.533      0.000      -1.398      -1.165
Stroke_Yes                    -0.9433      0.077    -12.204      0.000      -1.095      -0.792
Services_MRI                  -0.6170      0.163     -3.780      0.000      -0.937      -0.297
Services_CT Scan              -0.5528      0.095     -5.795      0.000      -0.740      -0.366
Gender_Nonbinary              -0.5330      0.213     -2.508      0.012      -0.950      -0.117
Anxiety_Yes                   -1.1224      0.066    -16.879      0.000      -1.253      -0.992
HighBlood_Yes                 -1.2295      0.062    -19.696      0.000      -1.352      -1.107
Diabetes_Yes                  -1.0632      0.070    -15.130      0.000      -1.201      -0.925
Hyperlipidemia_Yes            -1.1044      0.065    -17.062      0.000      -1.231      -0.977
==============================================================================================
"""
```

Model Comparison

The initial model has a weighted avg precision score of .98, the same as what the reduced model shows. The initial model also has a recall and accuracy score of .98, both within 1% of what the reduced model scores as well.

The reduced model does maintain accuracy, precision and recall, but the Pseudo R square drops from 0.60 to 0.37, a significant decrease indicating a deprecation of model fit.

The accuracy scores and confusion matrix show that the reduced model will be able to consistently provide accurate predictions, even though the Pseudo R-Square does show that the initial model is a better fitted model to the data.

Reduced Model Metrics

```
Reduced Y Intercept          : [-16.87657473]
[[3.59422792e-02 9.64057721e-01]
 [9.99991553e-01 8.44719838e-06]
 [9.99999897e-01 1.03293971e-07]
 ...
 [4.02401272e-01 5.97598728e-01]
 [9.99981141e-01 1.88593356e-05]
 [5.95067708e-01 4.04932292e-01]]
              precision    recall  f1-score   support

           0       1.00      0.97      0.98      5040
           1       0.95      0.99      0.97      2960

    accuracy                           0.98      8000
   macro avg       0.97      0.98      0.98      8000
weighted avg       0.98      0.98      0.98      8000

[[4891  149]
 [  23 2937]]
```

The reduced model boasts an accuracy score, a weighted average of recall, and a weighted average of precision, all at 0.98.

Regression Equation

Y (ReAdmission) = 4.74 * Initial_days ,1.28*Initial_admin_Emergency Admission , 0.94 * Stroke_Yes , 0.61 * Services_MRI , 0.55 *Services_CT Scan , 0.53*Gender_Nonbinary , 1.12*Anxiety_Yes , 1.22*HighBlood_Yes , 1.06*Diabetes_Yes , 1.10*Hyperlipidemia_Yes ,16.87

Coefficient Interpretation

The Initial days coefficient of 4.74 means that every additional day longer stayed in the hospital will increase the likelihood of readmission by 14.43% (exp 4.74 = 114.43) compared to the relative chance of readmission than a person who has not stayed that additional day, all else equal.

Initial admin as emergency admission has a coefficient of -1.28 (exp -1.28 =0.27), indicating that a patient who is admitted due to an emergency is 0.27 times as likely to be readmitted than a person who was not, all else equal.

Stroke yes has a coefficient of -0.943 (exp -0.943= 0.38) which indicates that a patient who has had a stroke is 0.38 times as likely to be readmitted than a patient who had not, all else equal.

Services MRI has a coefficient of -0.617(exp -0.617=0.53), which indicates that a patient who has had an MRI is 0.53 times as likely to be readmitted than a patient who had not, all else equal.

Services CT scan has a coefficient of -0.552 (exp -0.552=0.57), which indicates that a patient who has had a CT scan is 0.57 times as likely to be readmitted than a patient who had not, all else equal.

Gender nonbinary has a coefficient of -0.533 (exp -0.533=0.58), which indicates that a nonbinary gender patient is 0.58 times as likely to be readmitted than a patient who is not, all else equal.

Anxiety yes has a coefficient of -1.12 (exp -1.12= 0.32), which indicates that a patient who has anxiety is 0.32 times as likely to be readmitted than a patient who had not, all else equal.

High blood yes has a coefficient of -1.22 (exp -1.22=0.29) which indicates that a patient who experiences high blood pressure is 0.29 times as likely to be readmitted than a patient who does not, all else equal.

Diabetes yes has a coefficient of -1.06 (exp -1.06= 0.34) indicating a patient who has diabetes is 0.34 times as likely to be readmitted than a patient who does not, all else equal.

Hyperlipidemia has a coefficient of -1.10 (exp -1.10=0.33) indicating that a patient who has hyperlipidemia is 0.33 times as likely to be readmitted than a patient who does not, all else equal.

Statistical & Practical Significance

Practically, this can be used to give a very accurate estimate of whether readmission will be necessary, which will allow for more accurate preparation in staffing, equipment and supplies. This model could also be used to replace large amounts of stored data, as the readmission category will quite accurately be predicted based on the rest of the saved fields, that data could be erased and calculated as needed for cost decrease. Some companies may also be able to determine that the columns not used in any relevant models are not worth the cost of storing long term.

Limitations

All statistical models are built using past data, this model did not predict any of this data, it is created using the outcome. This means this is retrofitted, applying and fitting perfectly to past data, but we have no understanding of whether future data will follow any of the same trends. There are always unseen factors that the model cannot account for, such as natural disasters, spread of disease, or a change in the availability of some alternative form of care.

Additionally, the data itself is likely not perfect. Patients may not have a good understanding of what they are being asked, people may answer differently based on mood, recency bias, beliefs, etc. There are many potential ways datapoints could be inaccurate when analyzing a large population.

Course of Action

With the benefit of the reduced model, it is recommended that the hospital run a Readmission analysis on patients upon discharge from their initial stay. This will allow the hospital to then forecast staffing needs, equipment, medication, and the hospital should prepare accordingly. If desired, the hospital would also now have the option to remove the ReAdmission field from long-term data storage and use the reduced model to determine this value when necessary.